

Dhan Raj Lawati

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Professional Summary

Detail-oriented Ph.D. student in materials engineering with extensive hands-on experience in solid-state spin qubits, nitrogen-vacancy (NV) center-based magnetometry, and quantum sensing platforms to advance quantum technology from materials development through device integration. Proficient in designing and integrating quantum sensing systems including scanning NV microscopy, SQUID magnetometry, and cryogenic spin-state characterization targeting lower error rates, with hands-on experience in semiconductor processing and precision materials engineering for compact quantum device development. Proven ability to work independently and collaborate across multidisciplinary teams with strong communication skills, preparing formal reports and presenting findings in cross-functional settings. Passionate about leveraging expertise in quantum materials and device engineering to identify new growth areas utilizing state-of-the-art approach for bridging materials engineering with commercial quantum device scaling.

Skills and Competencies

- **NV-diamond and Quantum Photonics:** Scanning NV microscopy, Wide-field NV microscopy, Cryogenic NV microscopy, Nonlinear optics, Microwave engineering.
- **Fabrication, Processing and Characterization:** Semiconductor processing and compact quantum device fabrication using thin-film deposition, electron-beam lithography and etching; Superconducting Quantum Interference Device (SQUID), Atomic Force Microscopy (AFM), Magnetic Force Microscopy (MFM), Scanning Electron Microscopy (SEM), Transmission Electron Microscopy (TEM), XRD, Raman.
- **Computational & Data Analysis:** Python, MATLAB, Mathematica, LaTeX, LabVIEW, Origin, Gwyddion, VESTA, VASP; statistical modeling and multivariate analysis; DOE-guided workflow optimization; quantitative data interpretation.
- **Professional & Interpersonal Skills:** Detail-oriented, organized, and self-motivated; strong written and verbal communication; effective collaborator in multidisciplinary, high-performance research teams; proactive problem-solver with extensive experience troubleshooting complex experiments; adaptable, independent, and committed to rigorous scientific integrity and continuous improvement.

Education

Doctor of Philosophy in Materials Engineering

University of Nebraska- Lincoln, Lincoln, NE | CGPA: 3.90/4.00
Advisor: Prof. Abdelhane Laraoui

08/2024- Present

Master's in Physics

Tribhuvan University, Kirtipur, Nepal | CGPA: 3.87/4.00
Advisor: Prof. Rajendra Parajuli

12/2018- 12/2022

Experiences

Graduate Research Assistant: Quantum Sensing & Defect Discovery and Spectroscopy Lab

01/2025- Present

Project 1: High-sensitivity piezomagnetic imaging with NV quantum sensing platform

- Fabricated and characterized thin films using AFM, XRD, SEM, and SQUID, producing validated structural and magnetic datasets used for materials optimization and analysis.
- Designed and integrated a custom NV scanning microscopy system combining single-NV AFM probes, microwave excitation, and electromagnet bias fields, achieving up to 30% ODMR contrast and enabling reliable nanoscale magnetic sensing.
- Engineered multi-parameter experimental platforms for temperature (290–400 K), magnetic field-cooled (≤ 1 T), and strain-dependent studies to investigate piezomagnetic coupling phenomena in spintronic materials.
- Automated CW-ODMR imaging workflows in LabVIEW (fullB, isoB, Dual-isoB, Quenching), achieving 40 nm spatial resolution and $\sim 5 \mu\text{T}/\sqrt{\text{Hz}}$ sensitivity, producing high-resolution magnetic field maps.
- Built Python-based data processing pipelines for magnetic imaging reconstruction, statistical modeling, and visualization, generating quantitative magnetic domain analysis.

Project 2: Cryogenic magnetic sensing of 2D magnets with single electron spin in diamond

- Designed and fabricated microwave delivery loops via e-beam lithography to enable efficient spin-state excitation under cryogenic conditions, producing high-performance microwave structures for low-temperature NV experiments.
- Implemented and optimized a closed-cycle cryogenic system providing stable temperature control from 2–300 K, achieving $\sim 10\%$ ODMR contrast and enabling reliable quantum sensing measurements of 2D magnetic materials.
- Executed DC (ODMR, Rabi, Ramsey) and AC (T_1 , T_2 , XY8, CPMG) sensing protocols to quantify nanoscale stray magnetic fields and spin dynamics, generating coherence time measurements and magnetic noise spectra for low-temperature magnetic characterization.

Project 1: *Development of supercapacitor electrodes from bio-waste-derived nanomaterials for green energy applications*

- Synthesized biomass-derived activated carbon nanomaterials from bio-waste, producing sustainable, low-cost electrode materials with high surface area suitable for supercapacitor applications.
- Designed and fabricated binder-free working electrodes and performed electrochemical characterization using cyclic voltammetry (CV) and galvanostatic charge–discharge (GCD), achieving stable rectangular CV behavior and high specific capacitance, and generating validated electrochemical performance datasets.

Project 2: *Discovery and investigation of quantum materials using first-principles approach for optoelectronics applications*

- Designed and optimized density functional theory (DFT) simulation workflows, including structural relaxation, k-point sampling, and convergence testing, ensuring numerically stable and reproducible modeling of quantum materials.
- Performed first-principles electronic and optical property calculations (band structure, density of states, dielectric and absorption spectra) to predict material suitability for optoelectronic applications, generating quantitative structure–property datasets.

Leadership and Mentorship

Member - Mechanical and Materials Engineering Graduate Student Association (MME-GSA)

Spring 2026

- Served as active team member facilitating strategic planning and effective communication between the executive board and student members.
- Coordinated scheduling and logistics for meetings, seminars, and professional development events, promoting student engagement and participation in organizational activities.

Graduate Mentor

08/2025- Present

- Mentored two undergraduate researchers, resulting in two student-led symposium presentations and 100% retention in STEM pathways.

Conference and Symposium Judge

Spring 2025 & 2026

- Served as a graduate judge for undergraduate STEM poster presenters in university research days.
- Encouraged scientific communication and provided future work suggestions to participating students.

Publications

- **D. R. Lawati**, P. B. Karki, J. Kumar, *et al.*, “Spin Dynamics in the van der Waals Ferromagnet CrTe₂ Engineered by Nb Doping,” arxiv-2512.22058v1. DOI: <https://doi.org/10.48550/arXiv.2512.22058> (Submitted - Publication in process).
- J. Khan, **D. R. Lawati**, A. Dutta, F. N. Almutairi, and A. Ayari-Akkari, “Insight to structural, mechanical, electronic and optical properties of YSbPd and YSbPt half heusles: An ab-initio investigation,” *Optical and Quantum Electronics*, vol. 57, no. 2, pp. 1–16, 2025. DOI: <https://doi.org/10.1007/s11082-024-08020-z>.
- **D. R. Lawati**, J. Khan, A. Dutta, B. Akila, N. Algethami, and M. Al-Buriah, “Investigating structural and electronic features, band gap modulation, and optical absorption in ASnCl₃ (A = Na, K) perovskites for optoelectronic applications,” *Solid State Communications*, vol. 398, p. 115850, 2025. DOI: <https://doi.org/10.1016/j.ssc.2025.115850>.
- D. Behera, **D. R. Lawati**, M. Agouri, *et al.*, “A DFT insight into the physical features of alkaline based perovskite compounds AlnBr₃ (A = K, Rb),” *Solid State Ionics*, vol. 409, p. 116513, 2024. DOI: <https://doi.org/10.1016/j.ssi.2024.116513>.
- **D. R. Lawati**, P. Shrestha, S. Sharma, L. P. Joshi, and R. Parajuli, “A DFT study of the physical properties of novel vanadium-based half-heusler alloys VYAl (Y = Ni, Pd, and Pt) pertinent to their optoelectronics application,” *Physica Scripta*, vol. 99, p. 075997, 2024. DOI: <https://doi.org/10.1088/1402-4896/ad591c>.
- **D. R. Lawati**, H. K. Neupane, D. K. Chaudhary, *et al.*, “Structural, mechanical, electronic and optical properties of MgZnO₃ perovskite: First-principles study,” *Journal of Physics and Chemistry of Solids*, vol. 181, no. C, p. 111547, 2023. DOI: <https://doi.org/10.1016/j.jpcs.2023.111547>.

Achievements

- Received **Best Presentation Award**, International Conference on Lightning Electromagnetics and Applications of Semiconducting Materials (ICLEASM-2023).
- Received **Master Thesis Support Grant**, University Grants Commission, Nepal, Award no. MRS-77/78-S&T-121.
- Awarded with a **Full Merit-based Scholarship** for master’s study by Tribhuvan University.

Scientific Presentations

- **D. R. Lawati**, A. Erickson, Xiaoyu Yuan, *et al.*, “Study the effect of film thickness and substrate on antiferromagnetic domains in epitaxial Cr₂O₃ films using nitrogen-vacancy scanning probe microscopy,” American Physical Society (APS) Global Physics Summit, Denver, CO; March 15 - 20, 2026. (Oral)
- J. Kumar, **D. R. Lawati**, P. B. Karki, *et al.*, “Tuning the magnetic properties of Cr_{1-x}Nb_xTe₂ (x = 0 - 0.2) crystals by Nb doping,” American Physical Society (APS) Global Physics Summit, Denver, CO; March 15 - 20, 2026. (Oral)